Truth

Truth

SWAT4HCLS

AI and Big data in health

the dilemma of Truth

AI and Big data in health

the dilemma of Truth

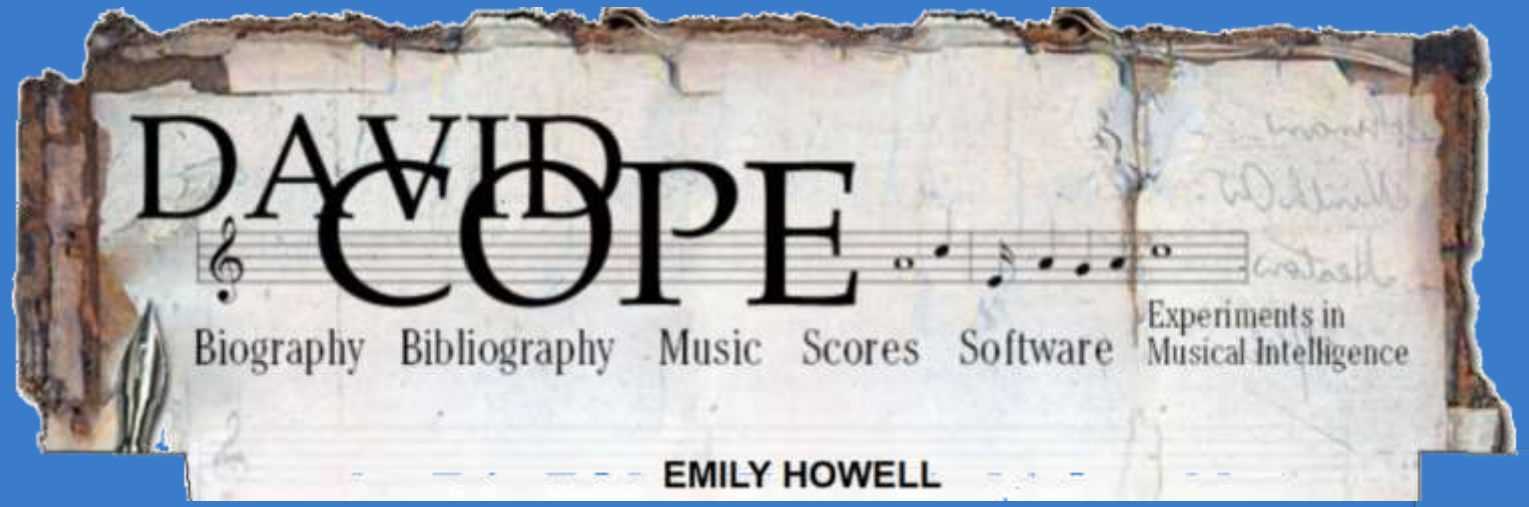


SWAT4HCLS

christian.lovis@hcuge.ch
medical information sciences

UNIVERSITÉ DE GENÈVE
FACULTÉ DE MÉDECINE

HUG Hôpitaux Universitaires Genève



Bach style chorale

Emmy Vivaldi

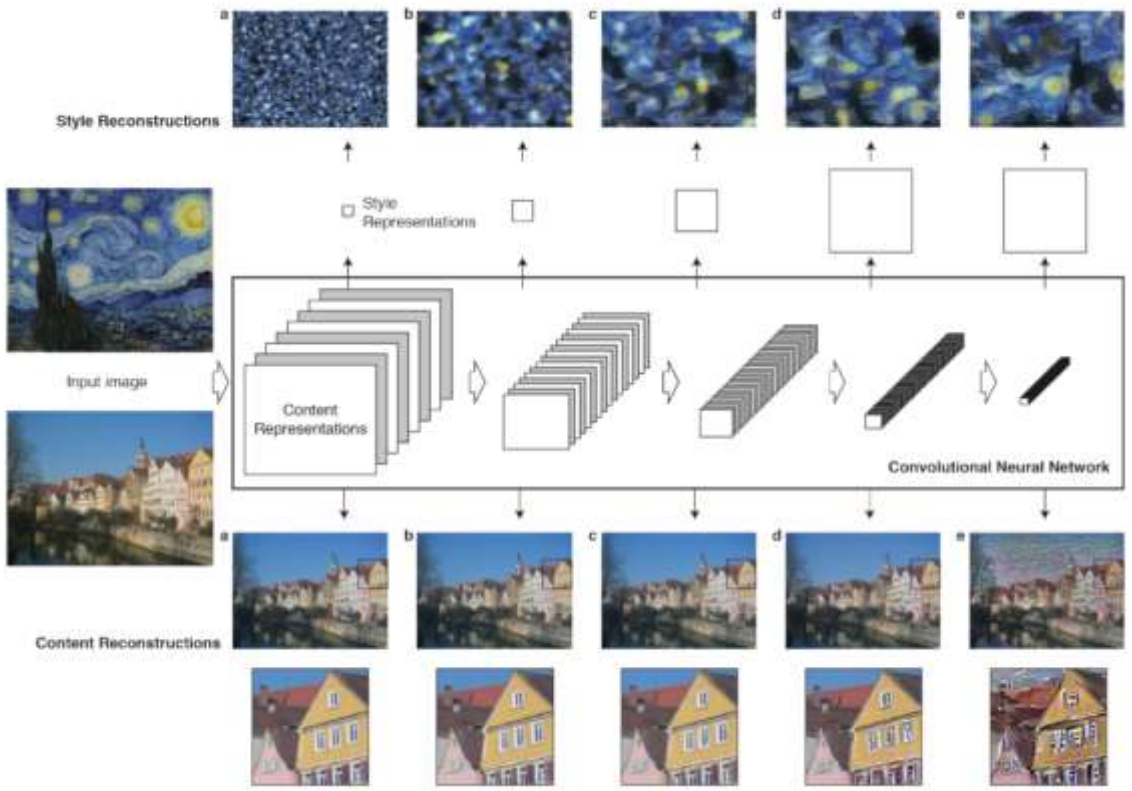
Emmy Beethoven beg 2

http://artsites.ucsc.edu/faculty/cope/Emily-howell.htm

AI in Musique

Experiences

David Cope, The Emily Howell project



convolutional neural network (CNN)

A Neural Algorithm of Artistic Style
Leon A. Gatys et coll, arXiv:1508.06576v1
The Bernstein Center for Computational Neuroscience, Germany





DOCTOR IS IN

Google is using 46 billion data points to predict the medical outcomes of hospital patients

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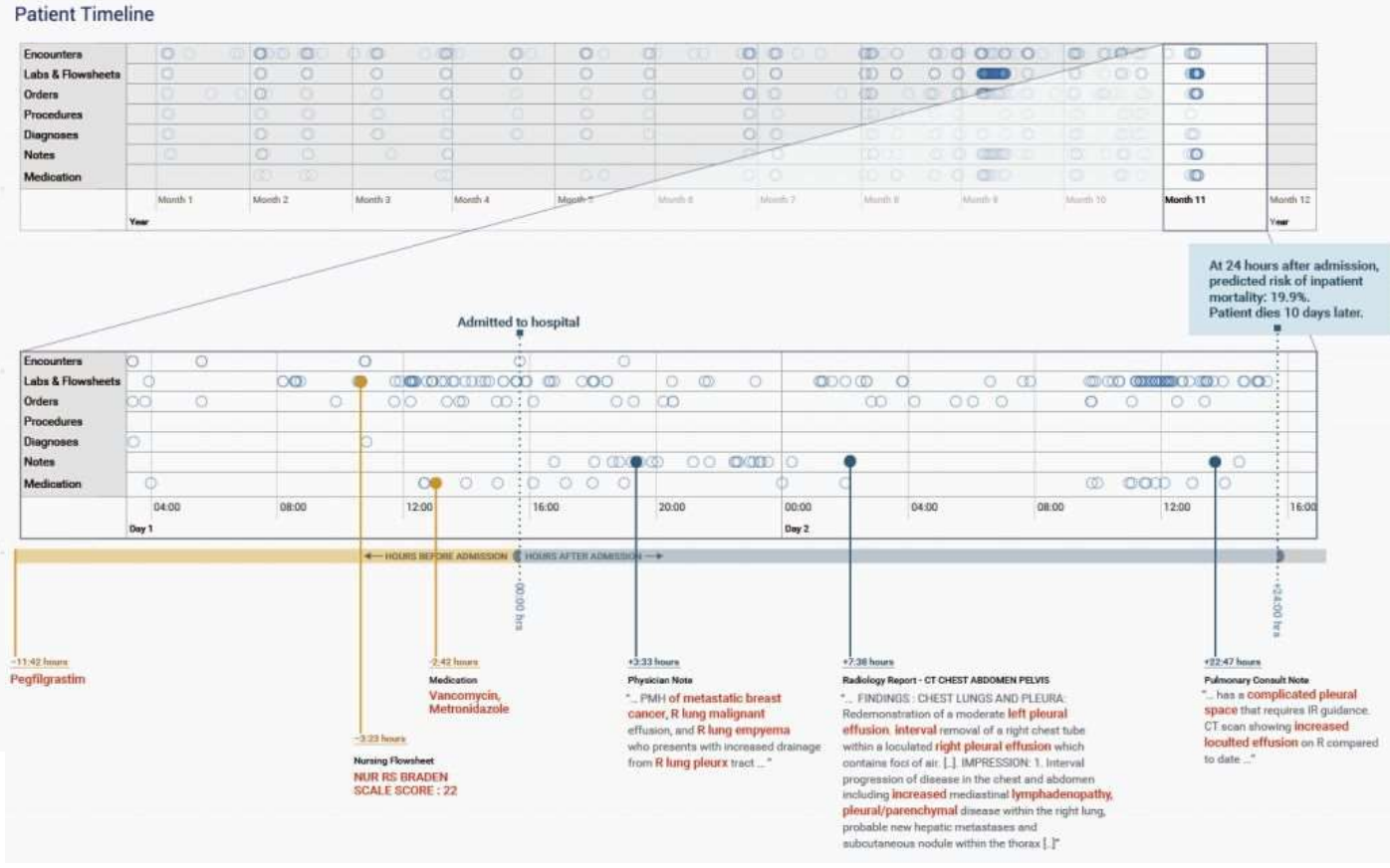
WRITTEN BY

Dave Gershgorn

OBSESSION

Machines with Brains

January 27, 2018



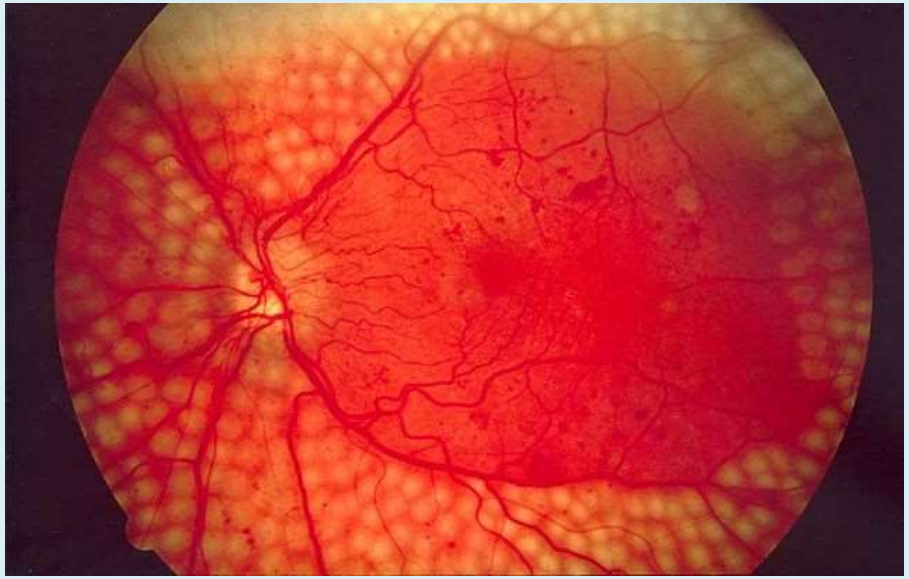
EDITORIAL

Artificial Intelligence With Deep Learning Technology Looks Into Diabetic Retinopathy Screening

Tien Yin Wong, MD, PhD; Neil M. Bressler, MD

Medium, severe retinopathy
Retinal oedema

Sensibility 87% - 90%
Specificity 98%



Retina for laser treatment when diabetic disease

128 175 images required

JAMA. 2016;316(22):2366-2367.
oi:10.1001/jama.2016.17563

THE LANCET Oncology

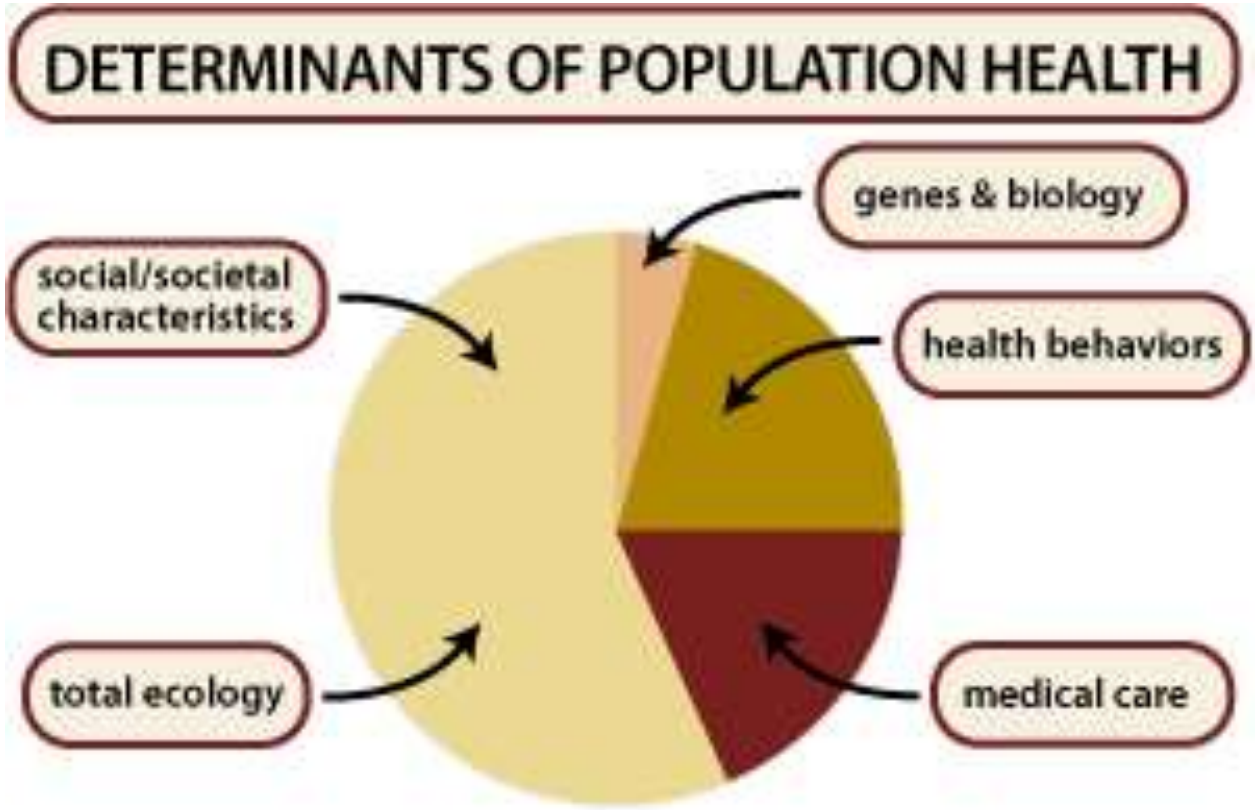
A radiomics approach to assess tumour-infiltrating CD8 cells and response to anti-PD-1 or anti-PD-L1 immunotherapy: an imaging biomarker, retrospective multicohort study
Sun R et al, DOI:https://doi.org/10.1016/S1470-2045(18)30413-3

Cancer : AI predicts response to immunotherapy

Nikos Paragios, co-author, prof in math

Only one out of 5 patients will have a good response to treatment, and one will suffer severe side effects, three will have no response and usual side effects.

https://www.sciencesetavenir.fr/high-tech/intelligence-artificielle/cancer-une-intelligence-artificielle-predit-les-chances-de-succes-de-l-immunotherapie_126965



Tarlov, A.R., Public Policy Frameworks for Improving Population Health. Annals of the New York Academy of Sciences, 1999. 896(SOCIOECONOMIC STATUS AND HEALTH IN INDUSTRIAL NATIONS: SOCIAL, PSYCHOLOGICAL, AND BIOLOGICAL PATHWAYS): p. 281-293.l


20:10
JEU
VEN
SAM
DIM
LUN
MAR
MER










caller id
text messages
meeting reminders




TITAN


Nest
3:15
Home
36 Willow Lane
70
Living Room


Withings

APPLE \ SCIENCE \ TECH

Why an Apple Watch with EKG matters

The FDA has approved the EKG feature, meaning the Watch can be used as a medical device

By Angela Chen | @chengela | Sep 12, 2018, 1:27pm EDT






20:50
Détails quotidiens
mars 20, 2017
Pas
5 001 | 100% d'objectif
Chens-sur-Léman Cyclisme
40:28 | 17.95 km
Genève Cyclisme
45:06 | 17.92 km
Sommeil
7h 45m
Fréq. cardiaque
48 bpm - 161 bpm
Poids
71.4 kg


21:36
Détails quotidiens
mars 24, 2017
Pas
3 636 | 72% d'objectif
Rameur d'intérieur
1:24:28 | –
Sommeil
9h 29m
Fréq. cardiaque
38 bpm - 181 bpm
Poids
71.4 kg


MATT DAMON
JODIE FOSTER
ELYSIUM
FROM THE DIRECTOR OF 'DISTRICT 9'
IMAX 08.09.13



Business Report

IBM Aims to Make Medical Expertise a Commodity

Big Blue thinks its *Jeopardy!* champion Watson can make money by offering health-care providers new expertise without hiring new staff.

by Tom Simonite July 21, 2014

engadget

For a dollar, an AI will examine your medical scan

Zebra-Med's tech helps radiologists find heart, liver, bone and other diseases.

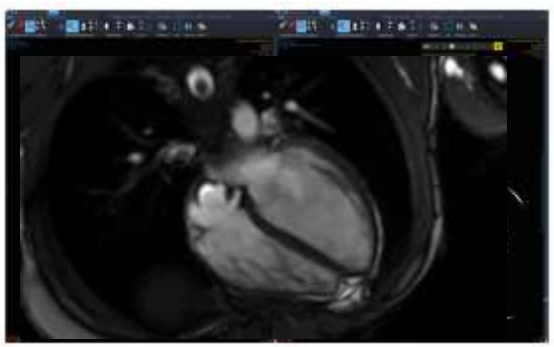


Steve Dent, @stevetdent
10.27.17 in Medicine

5
Comments

2473
Shares

A company called Zebra Medical Vision (Zebra-Med) has unveiled a new service called Zebra AI1 that uses algorithms to



examine your medical scans for a dollar each. The deep learning engine can examine CT, MRI and other scans and automatically detect lung, liver, heart and bone diseases.

New capabilities like lung and breast cancer, brain trauma, hypertension and others are "constantly being released," the company says. The results are then passed on to radiologists, saving them time in making a diagnosis or requesting further tests.

REALITY



UNEMPLOYED

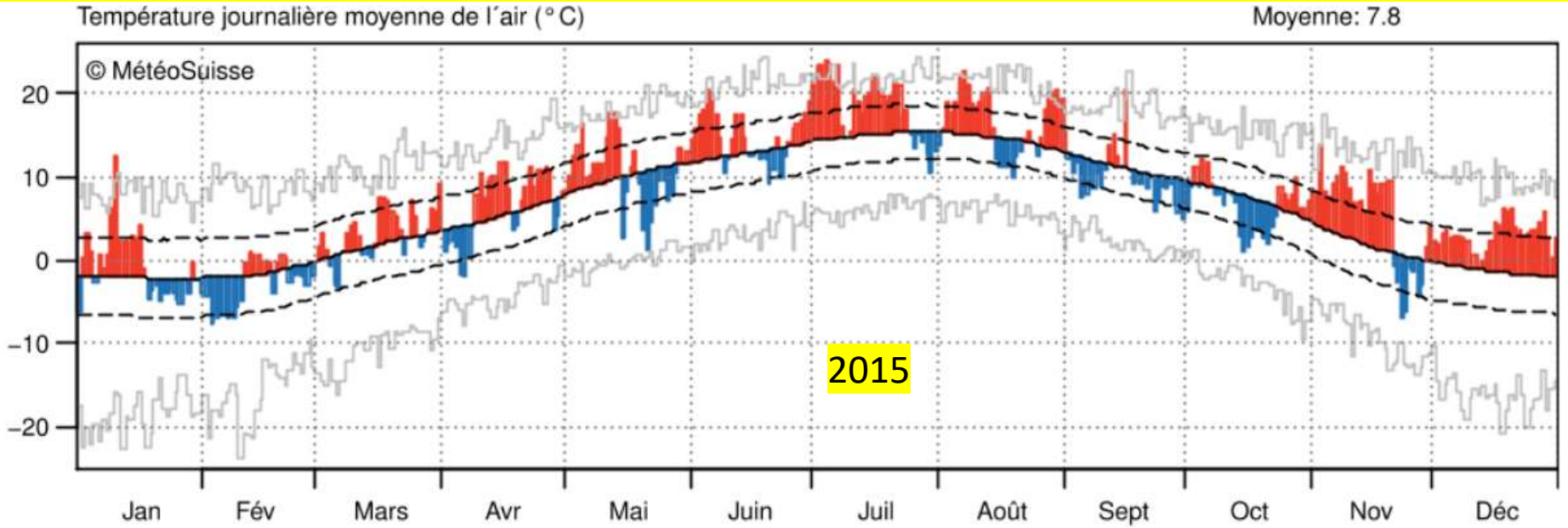
VS.

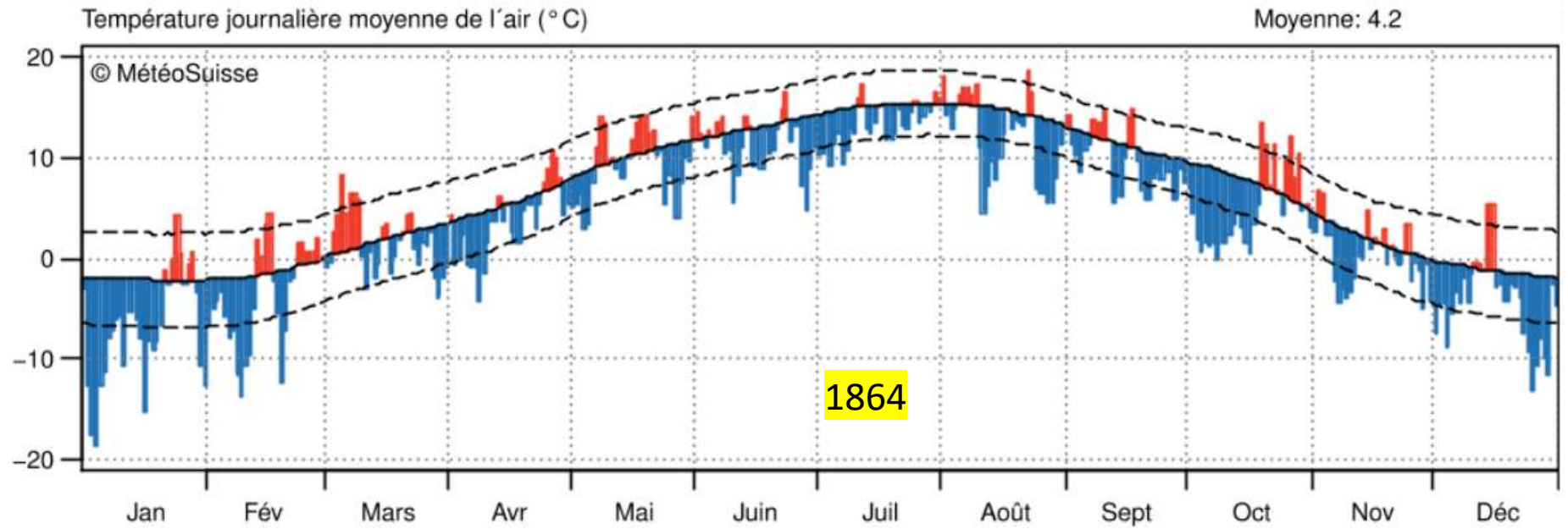
LinkedIn



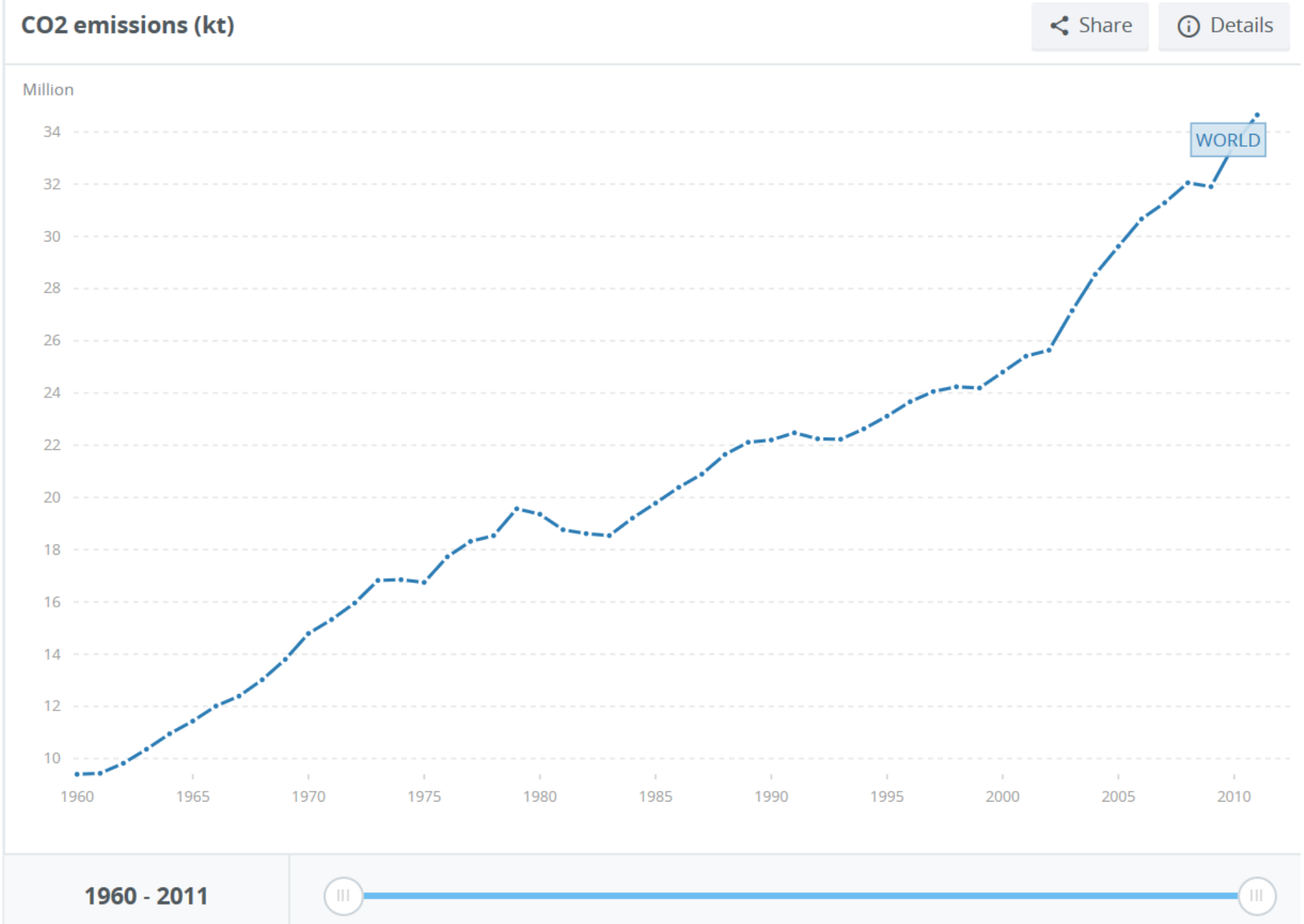
Blockchain Enthusiast | Cryptocurrency Evangelist | Influencer | Inspirer | Chief Visionary | Serial Entrepreneur (i.e. every business I started has failed) | Founder (Omission) | Philanthropist (Another Omission) | Empowering (Something) | Life Coach | Father | Trendsetter | Top 1% of LinkedIn Profiles (According to Myself) | Speaker | TEDx (2 x Attendee) | ICO Advisor |

Engelberg, 1036m, 1864-2015: + 3,5 °C





http://www.meteosuisse.admin.ch/home/climat/actuel/evolution-du-climat.html?region=Carte&station=eng



http://data.worldbank.org/topic/climate-change

Donald Trump On Climate Change: 'I Believe It Goes Up And It Goes Down'

"So I am not a believer, and I will, unless somebody can prove something to me, I believe there's weather."

09/22/2015 07:25 pm ET | **Updated** Sep 22, 2015

180



Philip Lewis

Fellow, The Huffington Post



ASSOCIATED PRESS

Senate votes that climate change is real

11.2K 43

COMMENTS 2885

By Laura Barron-Lopez - 01/21/15 04:48 PM EST



Central Dogma of Molecular Biology

by
FRANCIS CRICK
MRC Laboratory of Molecular Biology,
Hills Road,
Cambridge CB2 2QH

The central dogma of molecular biology deals with the detailed residue-by-residue transfer of sequential information. It states that such information cannot be transferred from protein to either protein or nucleic acid.

"The central dogma, enunciated by Crick in 1958 and the keystone of molecular biology ever since, is likely to prove a considerable over-simplification."

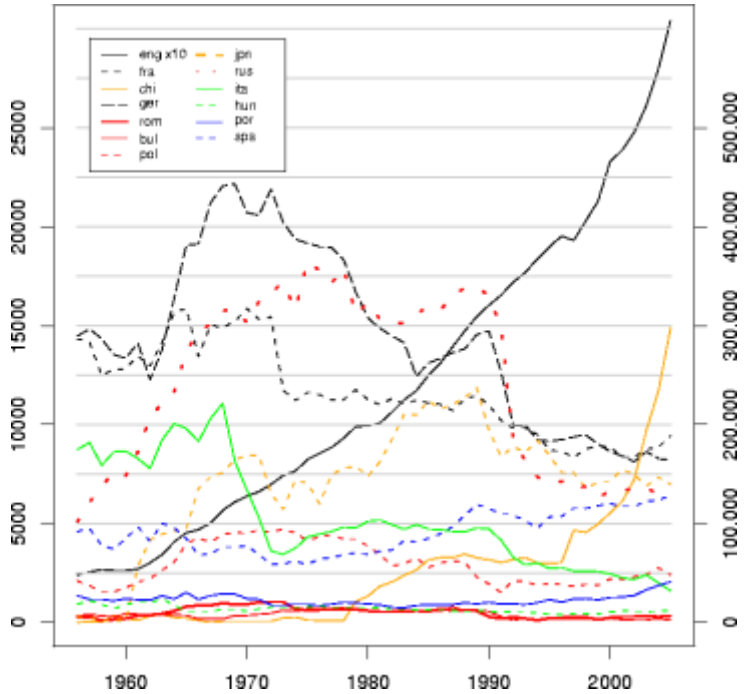
THIS quotation is taken from the beginning of an unsigned article[1] headed "Central dogma reversed", recounting the very important work of Dr Howard Temin[2] and others[3] showing that an RNA tumour virus can use viral RNA as a template for DNA synthesis. This is not the first time that the idea of the central dogma has been misunderstood, in one way or another. In this article I explain why the term was originally introduced, its true meaning, and state why I think that, properly understood, it is still an idea of fundamental importance.

The central dogma was put forward[4] at a period when much of what we now know in molecular genetics was not established. All we had to work on were certain fragmentary experimental results, themselves often rather uncertain and confused, and a boundless optimism that the basic concepts involved were rather simple and

analogous to thymine in DNA, thus giving four standard symbols for the components of nucleic acid.

The principal problem could then be stated as the formulation of the general rules for information transfer from one polymer with a defined alphabet to another. This could be compactly represented by the diagram of Fig. 1 (which was actually drawn at that time, though I am not sure that it was ever published) in which all possible simple transfers were represented by arrows. The arrows do not, of course, represent the flow of matter but the directional flow of detailed, residue-by-residue, sequence information from one polymer molecule to another.

Now if all possible transfers commonly occurred it would have been almost impossible to construct useful theories. Nevertheless, such theories were part of our everyday discussions. This was because it was being tacitly assumed that certain transfers could not occur. It occurred to me that it would be wise to state these



http://dan.corlan.net/medline-trend

PubMed

2016

2’796/day
5’623 journals
86’468 genes

2017

Science AAAS

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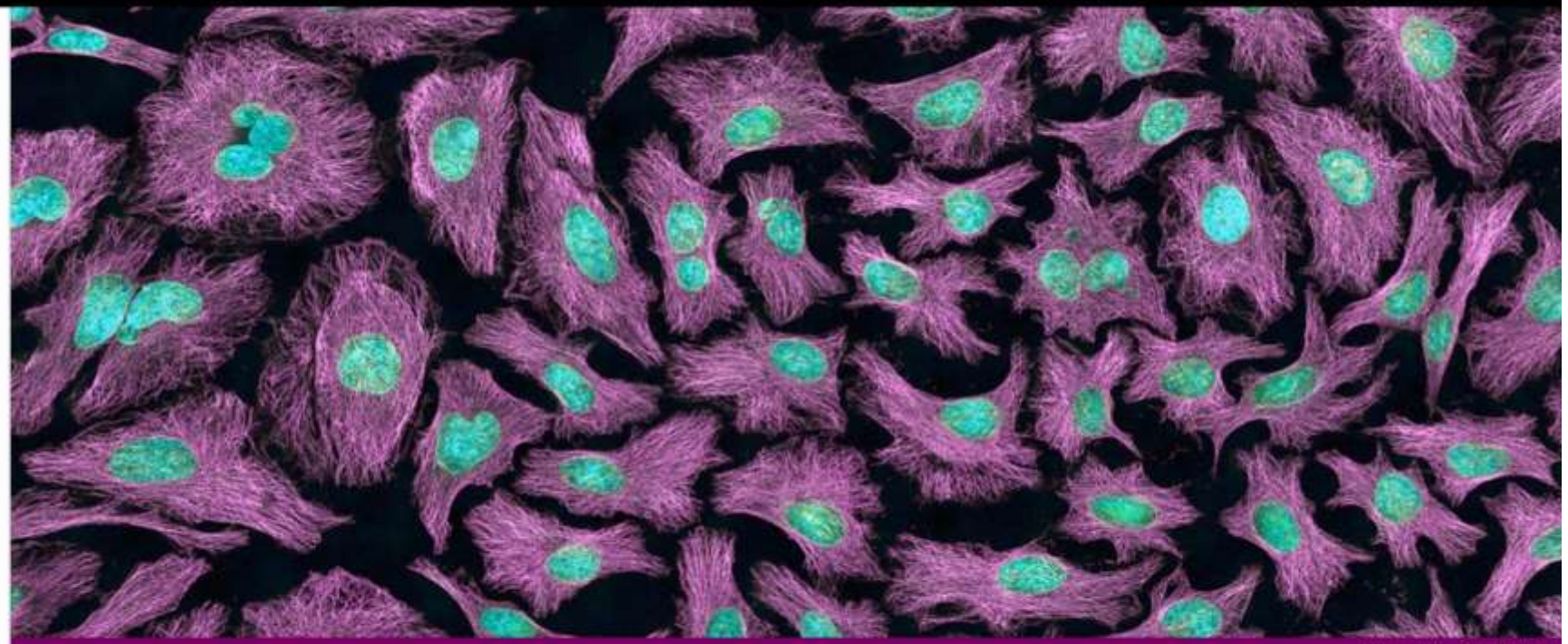


DAVIDE BONAZZI

Rigorous replication effort succeeds for just two of five cancer papers

By Jocelyn Kaiser | Jan. 18, 2017 , 1:00 PM

The first results of a high-profile effort to replicate influential papers in cancer biology are roiling the biomedical community. Of the five studies the project has tackled so far, some involving experimental treatments already in clinical trials, only two could be repeated; one could not, and technical problems stymied the remaining two replication efforts.



HeLa cells. Credit: NIH

Over 30,000 Published Studies Could Be Wrong Due to Contaminated Cells

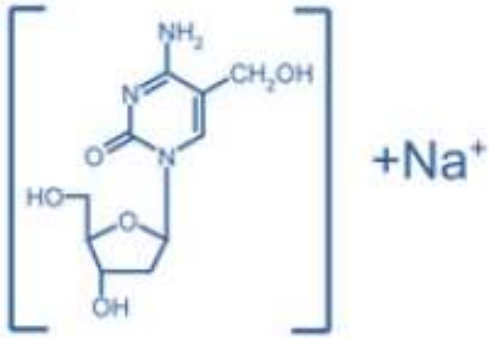
This is very, very bad.

PETER DOCKRILL 16 OCT 2017

Researchers warn that large parts of biomedical science could be invalid due to a cascading history of flawed data in a systemic failure going back decades.

A new investigation reveals more than 30,000 published scientific studies could be compromised by their use of misidentified cell lines, owing to so-called immortal cells contaminating other research cultures in the lab.

The Nuclear DNA Base 5-Hydroxymethylcytosine Is Present in Purkinje Neurons and the Brain



Skirmantas Kriaucionis and Nathaniel Heintz*

Despite the importance of epigenetic regulation in neurological disorders, little is known about neuronal chromatin. Cerebellar Purkinje neurons have large and euchromatic nuclei, whereas granule cell nuclei are small and have a more typical heterochromatin distribution. While comparing the abundance of 5-methylcytosine in Purkinje and granule cell nuclei, we detected the presence of an unusual DNA nucleotide. Using thin-layer chromatography, high-pressure liquid chromatography, and mass spectrometry, we identified the nucleotide as 5-hydroxymethyl-2′-deoxycytidine (hmdC). hmdC constitutes 0.6% of total nucleotides in Purkinje cells, 0.2% in granule cells, and is not present in cancer cell lines. hmdC is a constituent of nuclear DNA that is highly abundant in the brain, suggesting a role in epigenetic control of neuronal function.

we detected the presence of an unusual DNA nucleotide

Science
AAAS

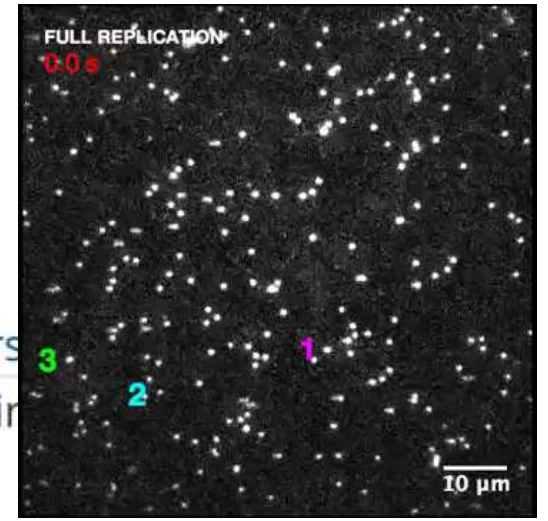
The Nuclear DNA Base 5-Hydroxymethylcytosine Is Present in Purkinje Neurons and the Brain
Skirmantas Kriaucionis and Nathaniel Heintz
Science **324**, 929 (2009);
DOI: 10.1126/science.1169786

PUBLIC RELEASE: 15-JUN-2017

DNA Replication Has Been Filmed For The First Time, And It's Not What We Expected

"It undermines a great deal of what's in the textbooks."

BEC CREW 19 JUN 2017



Here's proof of how far we've come in science - in a world-firs
recorded up-close footage of a single DNA molecule replicatir
questions about how we assumed the process played out.

The real-time footage has revealed that this fundamental part of life incorporates an unexpected amount of 'randomness', and it could force a major rethink into how genetic replication occurs without mutations.

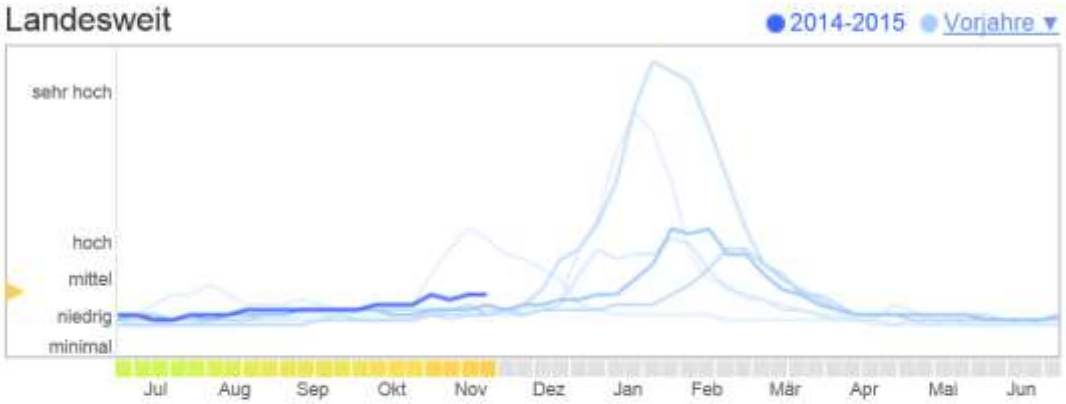
https://www.eurekalert.org/pub_releases/2017-06/uoc--vio061317.phph

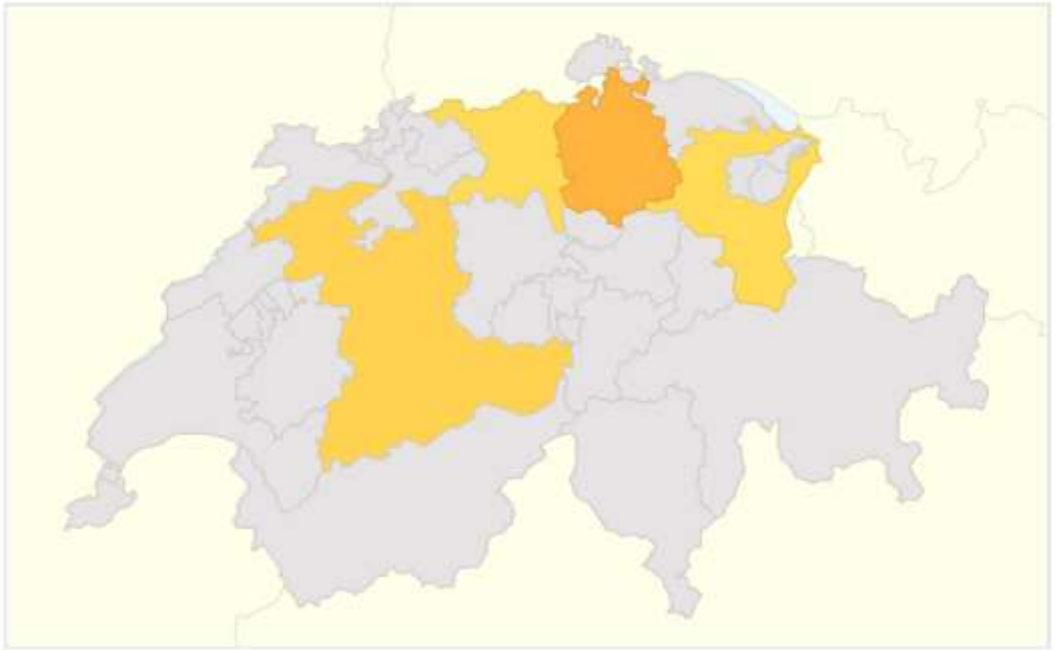
Grippe-Trends analysieren - Schweiz

Google hat festgestellt, dass die Häufigkeit bestimmter Suchbegriffe Anhaltspunkt für die Häufigkeit von Grippefällen sein kann. Für die Google Grippe-Trends werden Daten der Google-Suche gesammelt und ausgewertet. Auf Grundlage der Ergebnisse wird anschließend die Häufigkeit von Grippefällen geschätzt. Weitere Informationen »



Landesweit





nature International weekly journal of science

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Archive > Volume 494 > Issue 7436 > News > Article

NATURE | NEWS

عربي

When Google got flu wrong

US outbreak foxes a leading web-based method for tracking seasonal flu.

Declan Butler

13 February 2013

NewScientist Health

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SPACE TECH ENVIRONMENT HEALTH LIFE PHYSICS&MATH

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Google Flu Trends gets it wrong three years running

18:00 13 March 2014 by Hal Hodson

For similar stories, visit the Bird Flu Topic Guide

Google may be a master at data wrangling, but one of its products has been making bogus data-driven predictions. A study of Google's much-hyped flu tracker has consistently overestimated flu cases in the US for years. It's a failure that highlights the danger of relying on big data technologies.

ds, which launched in 2008, monitors web searches across the



https://www.youtube.com/watch?v=CxAxRCv9WoA

The NEW ENGLAND JOURNAL of MEDICINE

Messerli FH.

Chocolate consumption, cognitive function, and Nobel laureates.

N Engl J Med. 2012
Oct 18 367(16):1562-4.

PubMed PMID: 23050509h

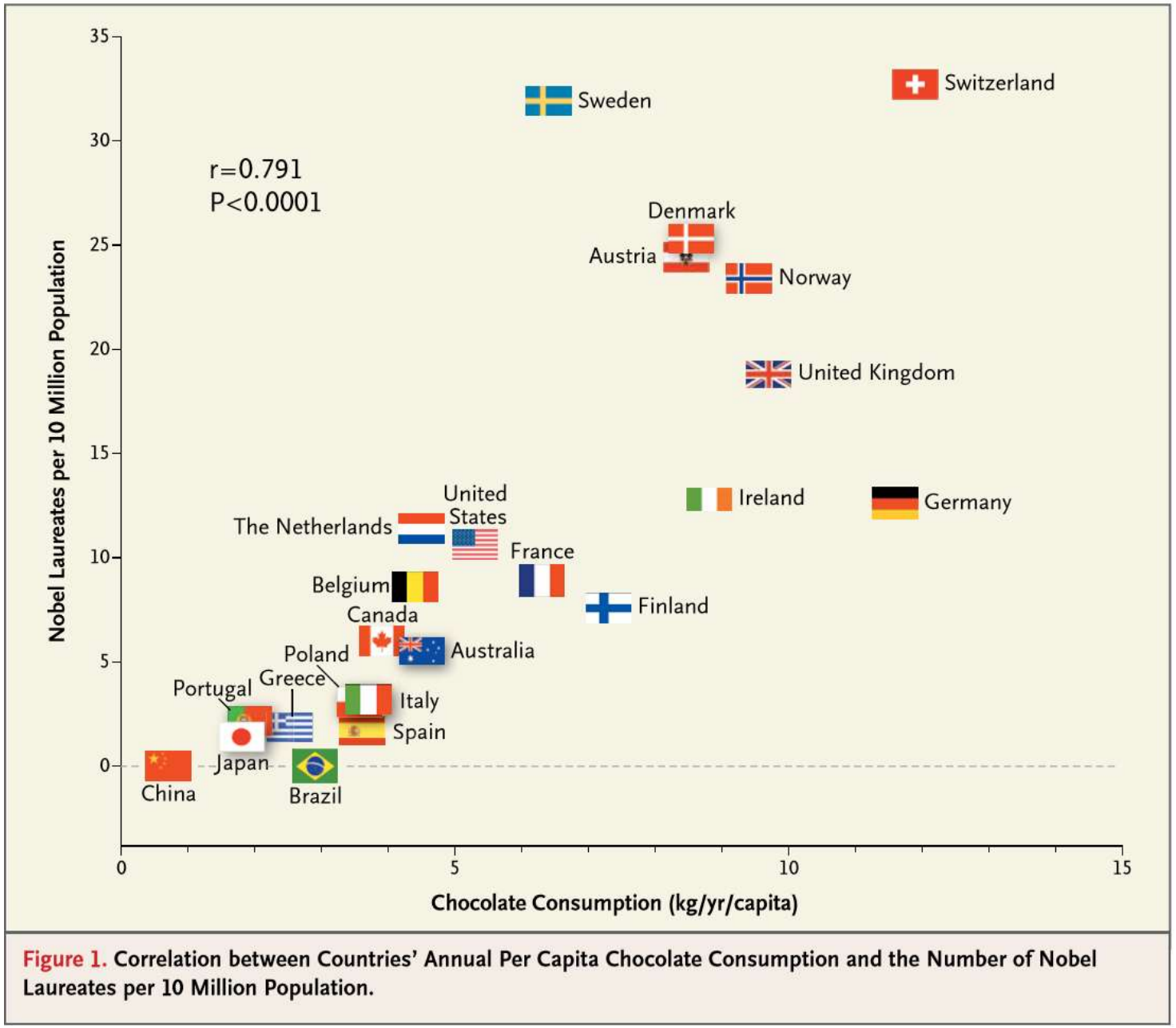


Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.



Effect on PPV of prevalence on a test with sensitivity of 99 % and specificity of 95 %				
	Prevalence	**1%**	**10%**	**20%**
a	# persons in population	1,000	1,000	1,000
b	With condition	10	100	200
c	Without condition	990	900	800
d	True positive (b x 0.99)	10	99	198
e	False positive (c x (1-0.95))	50	45	40
f	Total positive (d + e)	60	144	238
	PPV (d / f)	**17%**	**69%**	**83%**

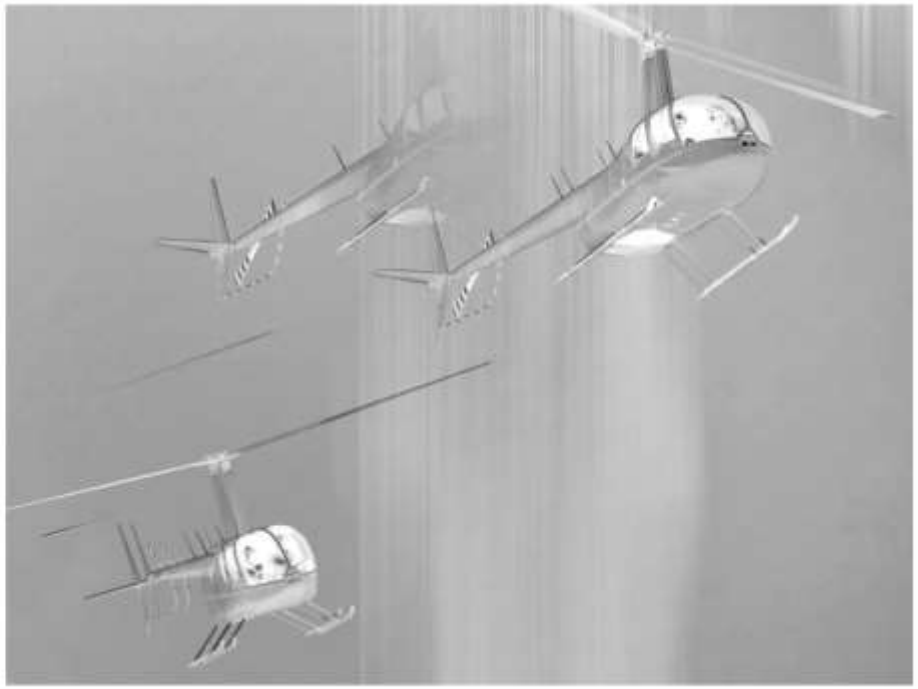
Chan Shah: Public health and preventive medicine in Canada.
Elsevier, Canada, 2003


Place de Cornavin
Genève
Google, Inc.
Street View – août 2014
26.05.2018
Les Brasseurs
METS DE BRASSERIE
FONDUES
Sansui
PÂQUIS
Hôtel Cristal 3*
assbindhotels.ch
Google

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TWEET
COMMENT
EMAIL

LOUISE MATSAKIS SECURITY 12.20.17 12:07 PM

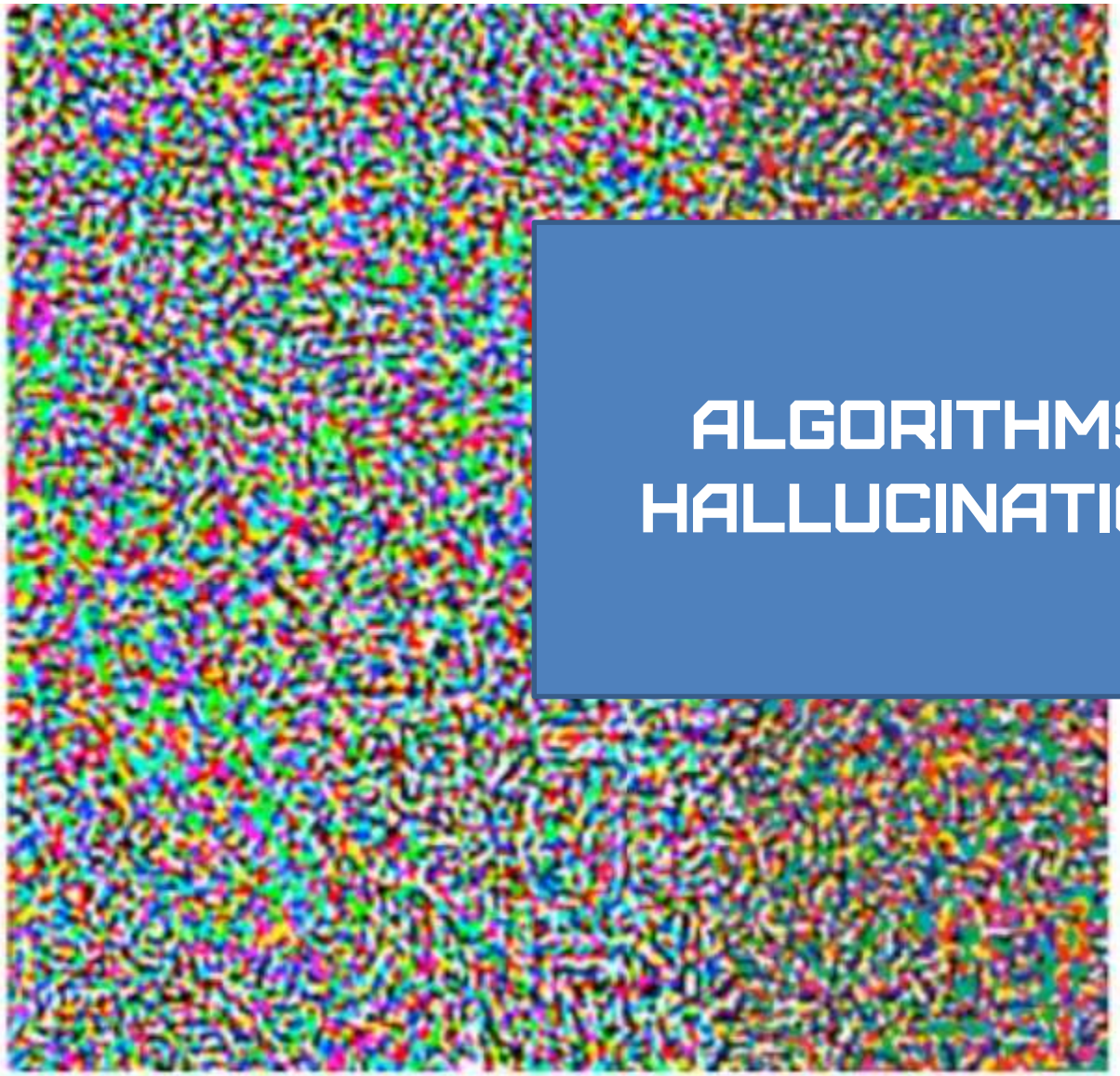
RESEARCHERS FOOLED A GOOGLE AI INTO THINKING A RIFLE WAS A HELICOPTER



https://www.wired.com/story/researcher-fooled-a-google-ai-into-thinking-a-rifle-was-a-helicopter/

[quote]

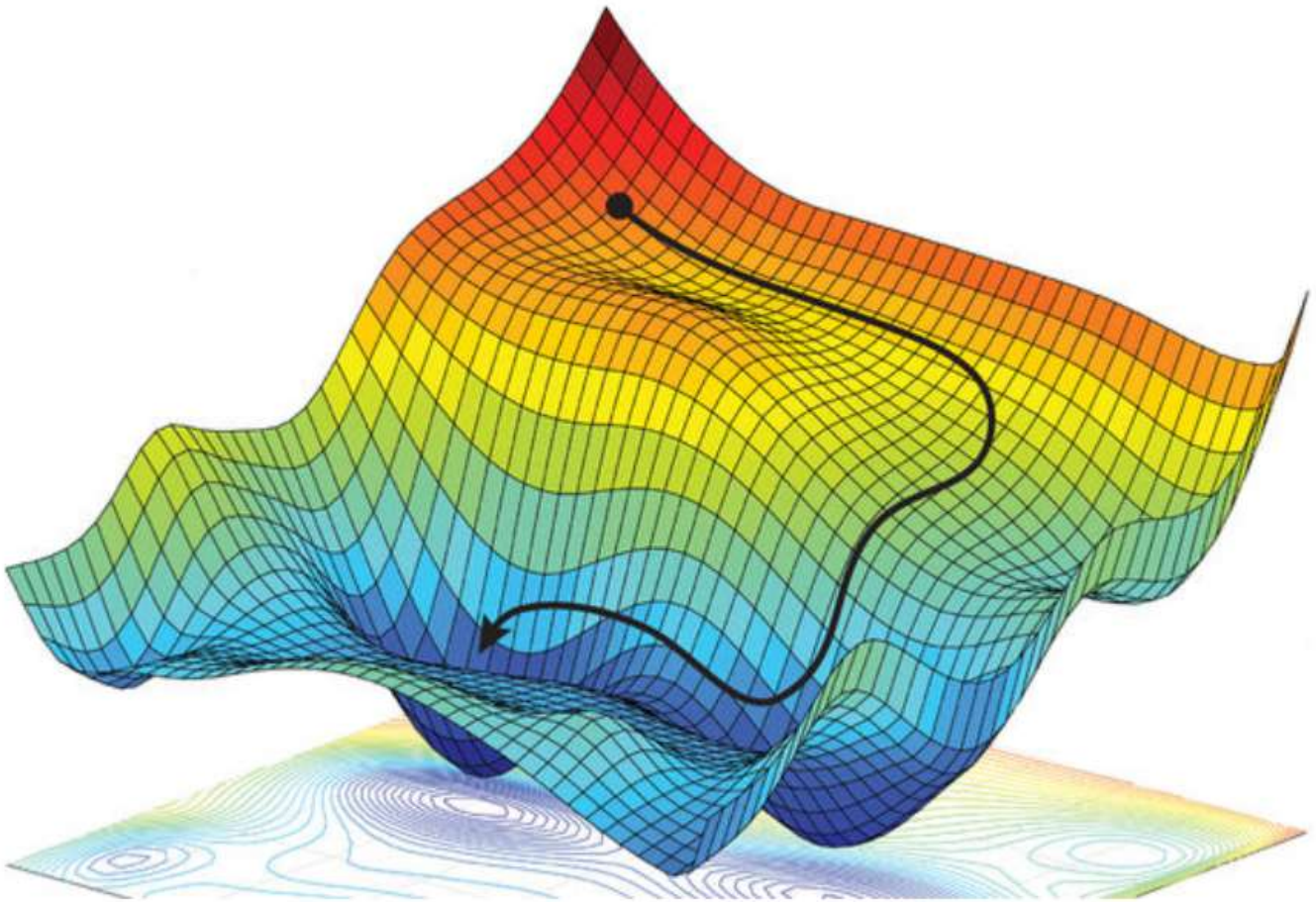
"in November another team at MIT (with many of the same researchers) published a study demonstrating how Google's InceptionV3 image classifier could be duped into thinking that a 3-D-printed turtle was a rifle."



ALGORITHMS HALLUCINATION



SHARE



Gradient descent relies on trial and error to optimize an algorithm, aiming for minima in a 3D landscape.
ALEXANDER AMINI, DANIELA RUS. MASSACHUSETTS INSTITUTE OF TECHNOLOGY, ADAPTED BY M. ATAROD/*SCIENCE*

AI researchers allege that machine learning is alchemy

By **Matthew Hutson** | May. 3, 2018 , 11:15 AM

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0

IN DEPTH | COMPUTER SCIENCE

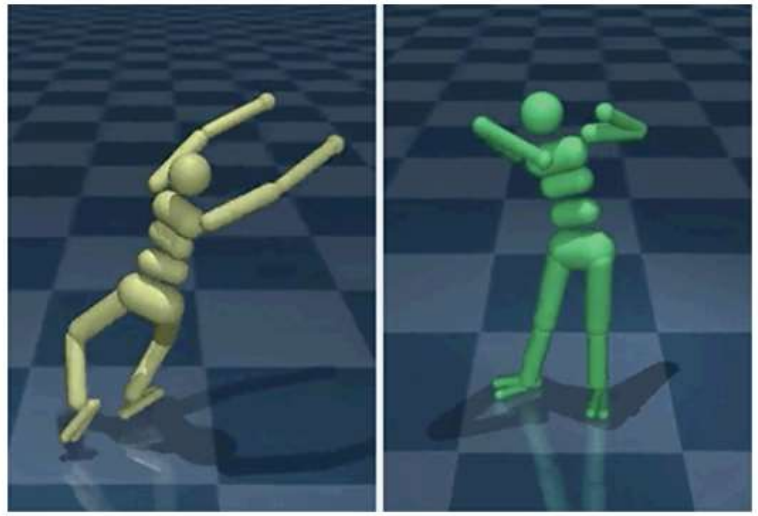
Artificial intelligence faces reproducibility crisis

Matthew Hutson

+ See all authors and affiliations

Science 16 Feb 2018:
Vol. 359, Issue 6377, pp. 725-726
DOI: 10.1126/science.359.6377.725

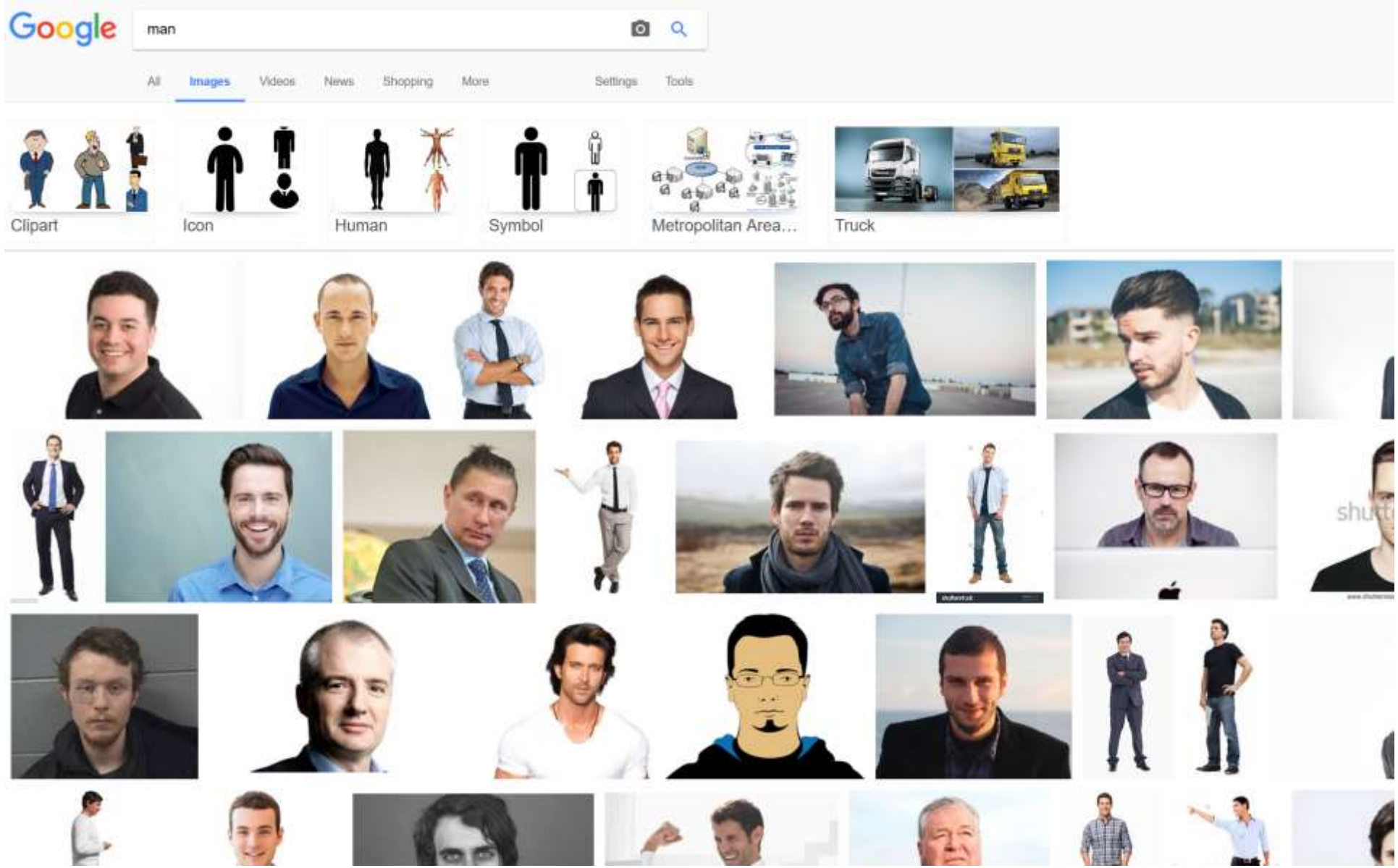
Last year, computer scientists at the University of Montreal (U of M) in Canada were eager to show off a new speech recognition algorithm, and they wanted to compare it to a benchmark, an algorithm from a well-known scientist. The only problem: The benchmark's source code wasn't published. The researchers had to recreate it from the published description. But they couldn't get their version to match the benchmark's claimed performance, says Nan Rosemary Ke, a Ph.D. student in the U of M lab. “We tried for 2 months and we couldn't get anywhere close.”

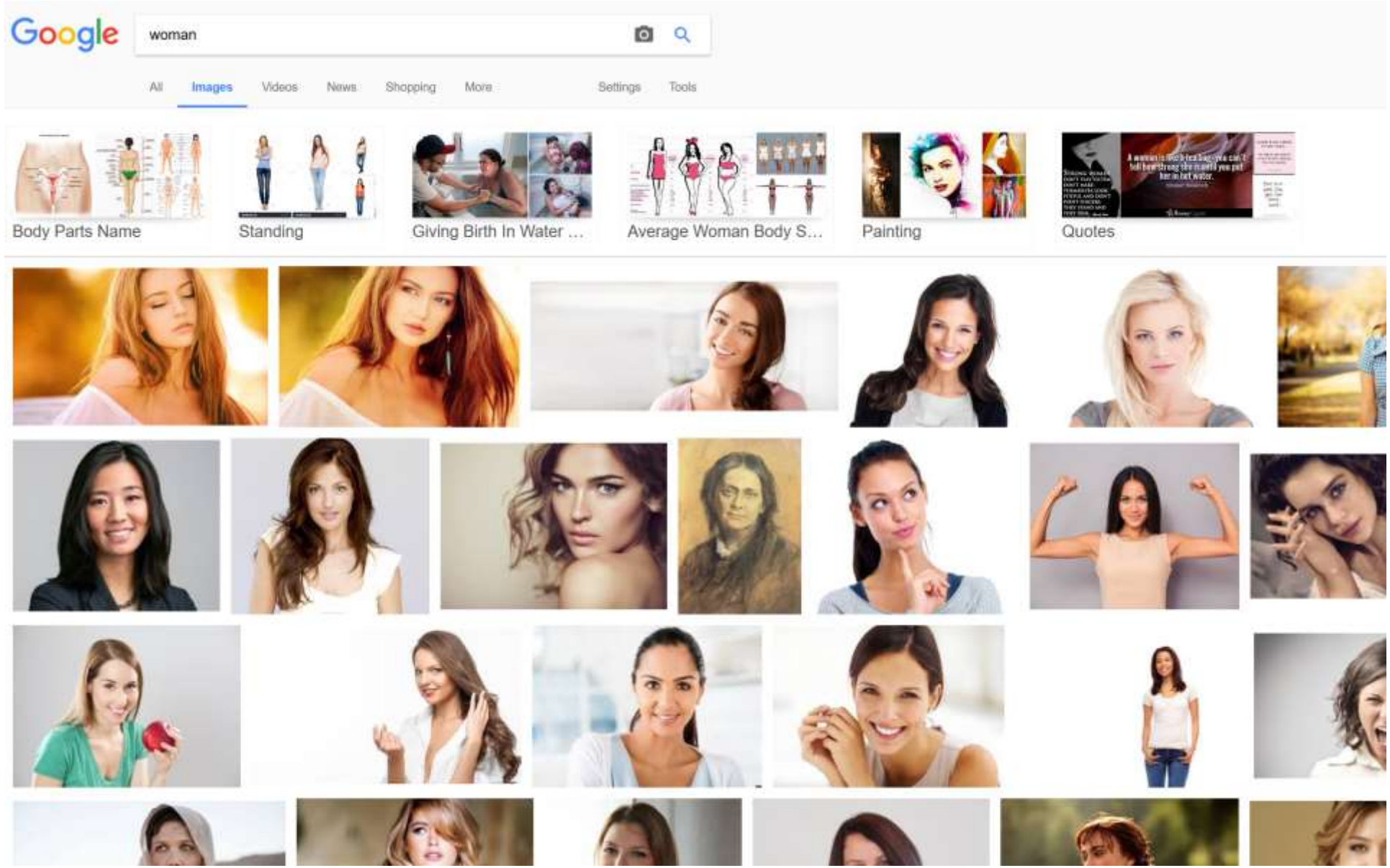


The same algorithm can learn to walk in wildly different ways. IMAGES: YUVAL TASSA



14 July 2018
Volume 392, Issue 10142


Google
man
All
Images
Videos
News
Shopping
More
Settings
Tools
Clipart
Icon
Human
Symbol
Metropolitan Area…
Truck


Google
woman
All
Images
Videos
News
Shopping
More
Settings
Tools
Body Parts Name
Standing
Giving Birth In Water ...
Average Woman Body S...
Painting
Quotes

Artificial intelligence (AI)

A beauty contest was judged by AI and the robots didn't like dark skin

The first international beauty contest decided by an algorithm has sparked controversy after the results revealed one glaring factor linking the winners



One expert says the results offer 'the perfect illustration of the problem' with machine bias. Photograph: Fabrizio Bensch/Reuters

This article is 7 months old

19,437 685

Sam Levin in San Francisco

@SamTLevin

email

Thursday 8 September 2016 23.42 BST

The first international beauty contest judged by "machines" was supposed to use objective factors such as facial symmetry and wrinkles to identify the most attractive contestants. After Beauty.AI launched this year, roughly 6,000 people from more than 100 countries submitted photos in the hopes that artificial intelligence, supported by complex algorithms, would determine that their faces

Reducing patient re-identification risk for laboratory results within research datasets

Ravi V Atreya,[1] Joshua C Smith,[1] Allison B McCoy,[2] Bradley Malin,[1,3] Randolph A Miller[1,4,5]

J Am Med Inform Assoc 2013;**20**:95–101. doi:10.1136/amiajnl-2012-001026

Table 1 Uniqueness for four, five, and six consecutive results of 10 representative laboratory tests and two panels (CBC and CHEM7)

Individual laboratory tests			
	Number of consecutive laboratory tests and proportion unique (with number of items qualifying for analysis)		
Test name	4	5	6
pH	0.590 (N=83 941)	0.937 (N=75 954)	0.994 (N=69 175)
Gluc	0.996 (N=133 259)	1.000 (N=110 669)	1.000 (N=93 693)
Ca	0.723 (N=51 905)	0.974 (N=41 140)	0.998 (N=33 283)
LymAbs	0.986 (N=27 591)	0.998 (N=22 361)	1.000 (N=18 615)
PCV	0.195 (N=201 941)	0.575 (N=172 619)	0.886 (N=149 514)
PT-INR	0.343 (N=47 768)	0.559 (N=38 875)	0.725 (N=32 188)
Chol	1.000 (N=780)	1.000 (N=613)	1.000 (N=478)
SGPT	0.996 (N=12 655)	0.999 (N=9807)	1.000 (N=7850)
CK	0.963 (N=6509)	0.979 (N=3659)	0.986 (N=2219)
Alb	0.649 (N=11 520)	0.924 (N=8606)	0.989 (N=6580)
Panel name	Laboratory panels		
CBC (five components)	0.988 (N=211 777)		
CHEM7 (seven components)	0.989 (N=239 253)		

Alb, albumin; Ca, calcium; CBC, complete blood count; CHEM7, blood test measuring electrolytes, glucose, and renal function; Chol, cholesterol; CK, creatine kinase; Gluc, glucose; LymAbs, absolute lymphocytes; PCV, hematocrit (packed cell volume); PT-INR, international normalized ratio for prothrombin time; SGPT, serum glutamic pyruvic transaminase.

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U.S. Department of Health and Human Services
Office for Civil Rights
Breach Portal: Notice to the Secretary of HHS Breach of Unsecured Protected Health Information

Under Investigation | Archive | Help for Consumers

As required by section 13402(e)(4) of the HITECH Act, the Secretary must post a list of breaches of unsecured protected health information affecting 500 or more individuals. The following breaches have been reported to the Secretary:

Cases Currently Under Investigation

This page lists all breaches reported within the last 24 months that are currently under investigation by the Office for Civil Rights.

Show Advanced Options

Breach Report Results

Expand All	Name of Covered Entity	State	Covered Entity Type	Individuals Affected	Breach Submission Date	Type of Breach	Location of Breached Information
	CA Department of Developmental Services	CA	Health Plan	582174	04/06/2018	Unauthorized Access/Disclosure	Paper/Films
	LifeBridge Health, Inc	MD	Healthcare Provider	538127	05/15/2018	Hacking/IT Incident	Network Server
	Oklahoma State University Center for Health Sciences	OK	Healthcare Provider	279865	01/05/2018	Hacking/IT Incident	Network Server
	St. Peter's Ambulatory Surgery Center LLC - d/b/a St. Peter's Surgery & Endoscopy Center	NY	Healthcare Provider	134512	02/28/2018	Hacking/IT Incident	Network Server
	Center for Orthopaedic Specialists - Providence Medical Institute (PMI)	CA	Healthcare Provider	81550	04/18/2018	Hacking/IT Incident	Network Server
	Tufts Associated Health Maintenance	MA	Health Plan	70320	02/16/2018	Unauthorized	Paper/Films

01.01.2018 – 11.06.2018 : 2 759 032 individuals affected

https://ocrportal.hhs.gov/ocr/breach/breach_report.jsf accessed 11.06.2018 – 08:45 CET

HEALTHCARE

Abundance of stolen healthcare records on dark web is causing a price collapse



SHARE

WRITTEN BY

Chris Bing

OCT 24, 2016 | CYBERSCOOP

The large amount of leaked patient records stolen and posted for sale to the dark web in recent months has caused prices for most of those records to drop, according to new research provided to CyberScoop and conducted by the Institute for Critical Infrastructure Technology and cybersecurity firms Flashpoint and Intel Security.

In the face of exceeding supply, stagnant demand and increased law enforcement attention, it's becoming increasingly difficult for criminals to make a living selling partial healthcare records, according to James Scott, a senior fellow at ICIT.

While the quality, quantity and sometimes origin of such electronic records will help dictate the price of any specific package for sale, average prices are largely trending downwards for individual, non-financial files, new research shows. The value of similar healthcare records that sold last year for roughly $75 to $100 dollars can now be found for around $20 to $50 dollars, Scott said.



UNINVITED GUESTS

COMMISSIONER	THINGTANK
YEAR	2015
TEAM	ANAB JAIN, JON ARDERN, JONATHAN FLINT, ALEXANDRA FRUHSTORFER
ACTOR	JAMES LEAHY

Thomas, aged 70, lives on his own after his wife died last year. His children send him smart devices to track and monitor his diet, health and sleep from a distance.

http://superflux.in/index.php/work/uninvited-guests/#

